

Arabian Sea model validation

Field	Entry
Region	Arabian Sea LME_032 Regional workbook
Catch source	Sea Around Us frozen regional catch, 1950–2019; landings, discards and total catch. Appendix reference: 2019 landings. Source snapshot catalogued 3 September 2026. Catch files Taxon metadata
Selected article	Mohamed K. S., Zacharia P. U., Muthiah C., Abdurahiman K. P., Nayak T. H. (2008). Trophic Modelling of the Arabian Sea Ecosystem off Karnataka and Simulation of Fishery Yields. CMFRI Bulletin 51. No DOI recorded. Paper PDF Official source
Other known articles	Vivekanandan et al. (2003), Southwest Coast of India coastal fisheries model, 11 groups. Mohamed & Zacharia (2009), Karnataka Ecosim yield simulations. Mohamed (2024), northeastern Arabian Sea model off Maharashtra/Gujarat. Mohamed-2009 PDF Mohamed-2024 PDF Vivekanandan-2003 PDF
Selected model	32_1_Arabian_Sea_off_Karnataka_(2000). Baseline landings 1999–2001; 27,000 km ² Karnataka shelf and parts of slope, exploitation to 200 m. 24 source groups plus synthetic diet_import (ID 25). Model JSON Group coefficients
Other models from the same article	One Karnataka baseline Ecopath model is identified. Original and balanced diet tables are reconstruction states of that model; Ecosim effort/yield scenarios are simulations, not separately selected Ecopath models.
Selection rationale	Article — ? Model — ?
Model extraction	- Printed diet column sums 0.999–1.001 are normalized by their actual sums. - GS not reported; default runtime values are filled. - The paper specifies zero biomass accumulation; this replaces missing BA LIM solving for balance (extracted as 0, not filled during runtime). ○ RESEARCHER VERIFIED – this does not affect SPPR results. Source reconstruction Loaded state
GE diagnostics	OK No negative SPPR entries across the basal-source columns. rho_living = 0.58 b = 0.52 Detritus (ID 24): SPPR = 2.22 Sppr calculation is almost balanced (inflow=outflow). - Balanced once BA is allowed to divert from 0.

Field	Entry
TE diagnostics	WARN: near-zero efficiency in (34) marine mammals (GE=1.5%, EE=5%). No negative SPPR entries across the basal-source columns. $\rho_{\text{living}} = 0.67$ Detritus(ID 24): SPPR = 0.32. Sppr calculation is almost balanced (inflow=outflow). Balanced once BA is allowed to divert from 0.
SPPR calculation	Model-input is not mass-balanced. Using non-zero BA values balances it, and does not affect PPR results. DC rows sum to 1. Removed groups from calculation: - marine mammals (GE=1.5%, EE=5, TL=4.06, sppr_TE=495K, sppr_GE=14.4K)
Geographic fit	A. Region covered by study: approximately 0.65–0.69%. B. Study area covered by region: approximately 95–100%.
Temporal fit	Model baseline 1999–2001 is transferred over catch 1950–2019, including the 2019 reference. Fixed model coefficients and assumed category weights do not reconstruct changing annual ecosystems. Regional lineage weights use identified total catch 1950–2019, then transfer across individual years and bases. Catfish and flatfish proxy weights have substantial reporting geography, gear or period mismatches.
Other	
Open issues and next action	- find a model that better covers the LME
Review and reproducibility	Researcher name: Ido Carmel review date: 01/10/2026 MODEL VALIDATED

Taxon mapping and coverage

Reference: 2019 landings; all 430 catch labels, including zero catches. Independent all-taxon simple-chain PPR; no group filter.

24 source groups plus one synthetic import; 430 catch taxa.

Simple-chain PPR totals 514,724,666.461 t C in 2019.

37 taxa lack a saved TL/coefficient. All have zero 2019 landings and contribute zero annual PPR; no positive-catch contribution is unknown.

Simple-chain PPR uses the independent simple trophic chain method.

[Excel taxon appendix and descriptive Sources sheet](#)

Overall confidence	Taxa	Catch percentage	PPR percentage
High	145	34.2506%	27.8791%
Medium	205	48.5275%	64.2253%
Low	40	4.0707%	4.5692%
Very low	40	13.1512%	3.3265%
Unresolved	0	0.0000%	0.0000%

Overall confidence uses the weaker required membership or allocation component.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Taxonomic extension from named source representatives	Medium	51.2386%
Explicit source member or member of a source-listed genus	High	20.7837%
Broad label interpreted through its recorded fishery classification	Medium	7.4699%
Complete fit to an explicitly defined generic compartment	High	7.0191%
Supported habitat and feeding-guild extension	Medium	5.8126%
Partial or conflicting habitat and feeding-guild fit	Low	4.2734%
Explicit coarse-category approximation to compatible compartments	Very low	3.1959%
Closest available ecological analogue with a material mismatch	Very low	0.1306%
Accepted-name synonym of an explicit source member	High	0.0763%

Membership evidence: Bulletin 51 membership table and group accounts, accepted-name taxonomy, provider reporting meaning and cited ecological sources.

Allocation weight rules

Plain-language rule	Confidence	PPR percentage
One supported compartment receives the entire taxon	High	89.3358%
Identified regional catch composition assumed for the coarse taxon	Medium	6.1289%
Source-model landings proportions assumed for the catch category	Medium	4.2395%
Identified regional catch composition assumed for the coarse taxon	Low	0.2958%

Weights and assumptions: seven identified-lineage catch mixtures are retained or reconstructed. Other splits use eligible source-group landings proportions at Medium allocation confidence, with zero candidates retained and transfer assumptions explicit. Siluriformes and Pleuronectiformes retain lineage splits at Low allocation confidence because reporting geography, gears or periods differ.

Each percentage is the independent simple-chain PPR associated with taxa using that rule.

Unresolved taxa: none. 40 Very low decisions cover 13.1512% of landings and 3.3265% of PPR. They comprise coarse fish/invertebrate mixtures, unrepresented herbivorous or specialised reef feeders, deep/oceanic analogues and Scomber. Exact taxa, weights and reasons are flagged in the linked appendix.

Very low decisions

Affected taxa	Why confidence is very low
Marine fishes not identified; Perciformes; Marine pelagic fishes not identified; Marine groundfishes not identified	Unknown fish composition is approximated by compatible harvested bony-fish guilds, water-column pools or benthic pools. A demersal tag does not establish unidentified-fish composition; the legacy Perciformes label is broader than four demersal pools. Whole source-group catch proportions are assumed transferable across regions, years and catch bases.
Anguilliformes	Adult eels and morays are approximated by demersal carnivore pools. Eel larvae in Micro Nekton do not represent adult fishery catch; size, diet and depth mixture remain uncertain. Published model catch proportions are an assumed allocation.
Chondrichthyes	Sharks and rays fit two source pools, but common metadata also includes chimaeras, absent from the model. The two elasmobranch pools provide an approximation rather than complete containment; model catch proportions supply an assumed split.
Scombroidei	The provider label includes tunas, bonitos and billfishes. Tunas plus Large Pelagics approximate those taxa; billfishes are ecological extensions, not source members. Published model catch proportions supply an assumed split.
Gobiidae	Unknown goby species and feeding composition are approximated by small benthic carnivory, omnivory and water-column feeding in three source pools. Reef and specialised feeding remain unmatched; model catch proportions supply an assumed split.
Siganus canaliculatus; Siganidae; Siganus; Crenidens crenidens; Scaridae; Bolbometopon muricatum; Scarus persicus; Acanthurus dussumieri; Siganus sutor; Scarus ghobban; Acanthuridae	Benthic Omnivores is the closest available bottom-feeding approximation. The model lacks a herbivorous reef-fish pool; specialised algal, seagrass or coral feeding differs materially from its sole and squilla diet.

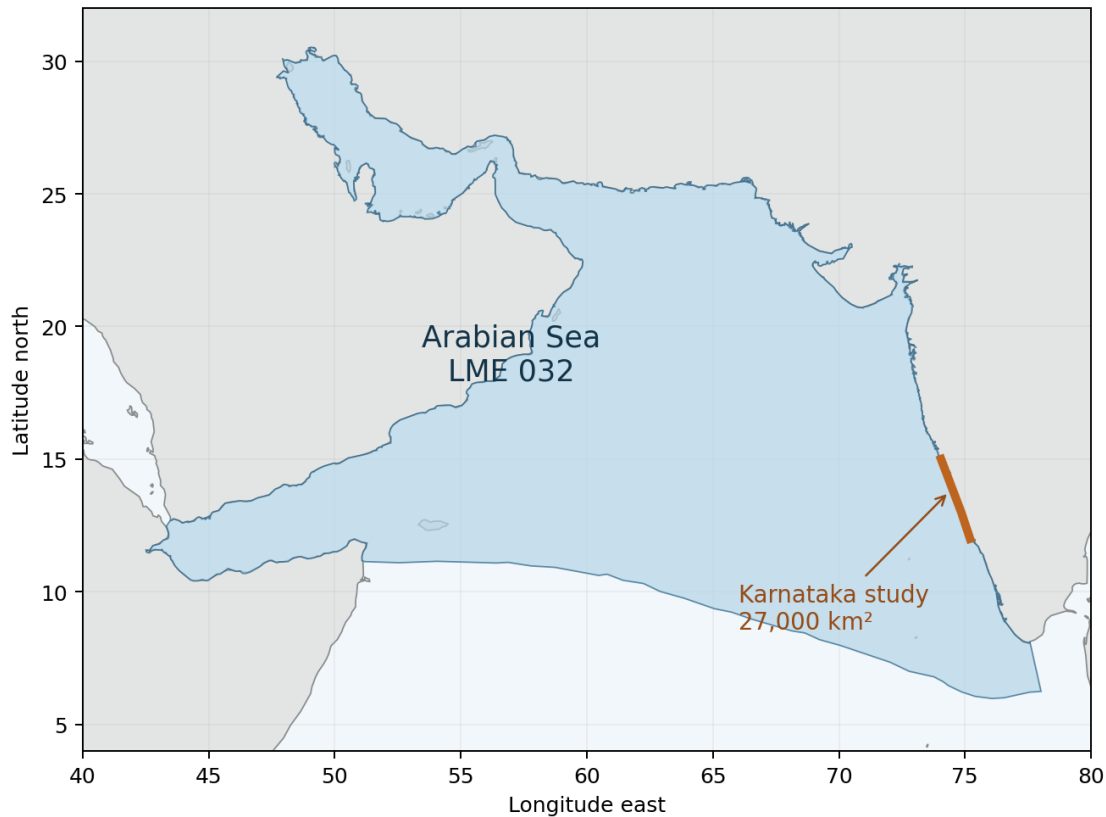
Affected taxa	Why confidence is very low
Pomacanthus maculosus; Platax orbicularis; Ehippidae; Platax; Chaetodontidae; Pomacentridae; Pomacanthidae	Benthic Omnivores is the closest available reef or bottom-feeding approximation. Specialised coral and sponge feeding, or mixed reef plankton and algal feeding, is absent from the model; direct source membership is not established.
Bramidae; Berycidae; Beryx; Brama brama; Gadiformes; Gempylidae; Lampris guttatus; Lepidocybium flavobrunneum; Ruvettus pretiosus	Large Benthopelagics, represented by ribbonfish and carangids, is the closest available water-column predator approximation. Deep or oceanic feeding and habitat, or coarse gadiform composition, are poorly represented by this shelf model exploited to 200 m.
Mollusca	Common metadata includes clams, seasnails, squids and octopuses. Cephalopods and Heterotrophic Benthos form a coarse approximation under an adult-fishery-catch assumption; larval zooplankton is excluded. Model catch proportions supply an assumed split.
Miscellaneous aquatic invertebrates	Unidentified demersal invertebrates may span molluscs, crustaceans, epifauna and infauna. Compatible adult pools provide an explicit coarse approximation; fish-bearing pools and pelagic larvae are excluded. Model catch proportions supply an assumed split.
Malacostraca	Common metadata includes lobsters, crabs, shrimps and krill. Harvested decapods, squilla, epifaunal crustaceans and large zooplankton approximate this mixture; micro-zooplankton eggs and larvae are excluded under an adult-catch assumption. Model catch proportions supply an assumed split.
Labridae	Wrasses vary in size and include benthic invertebrate feeders and planktivores. Medium and small benthic feeders plus the small water-column pool provide an approximation; specialised cleaning and reef habitats remain unmatched. Model catch proportions supply an assumed split.
Scomber	The dedicated Rastrelliger kanagurta pool is the closest available mackerel analogue. Scomber is not listed among source members and can differ materially in feeding and habitat.

Geographic evidence

Target region R and Karnataka study area S. Approximate overlap uses the published 27,000 km² footprint and target boundary; the study's coordinate rectangle is not treated as its area.

Region area

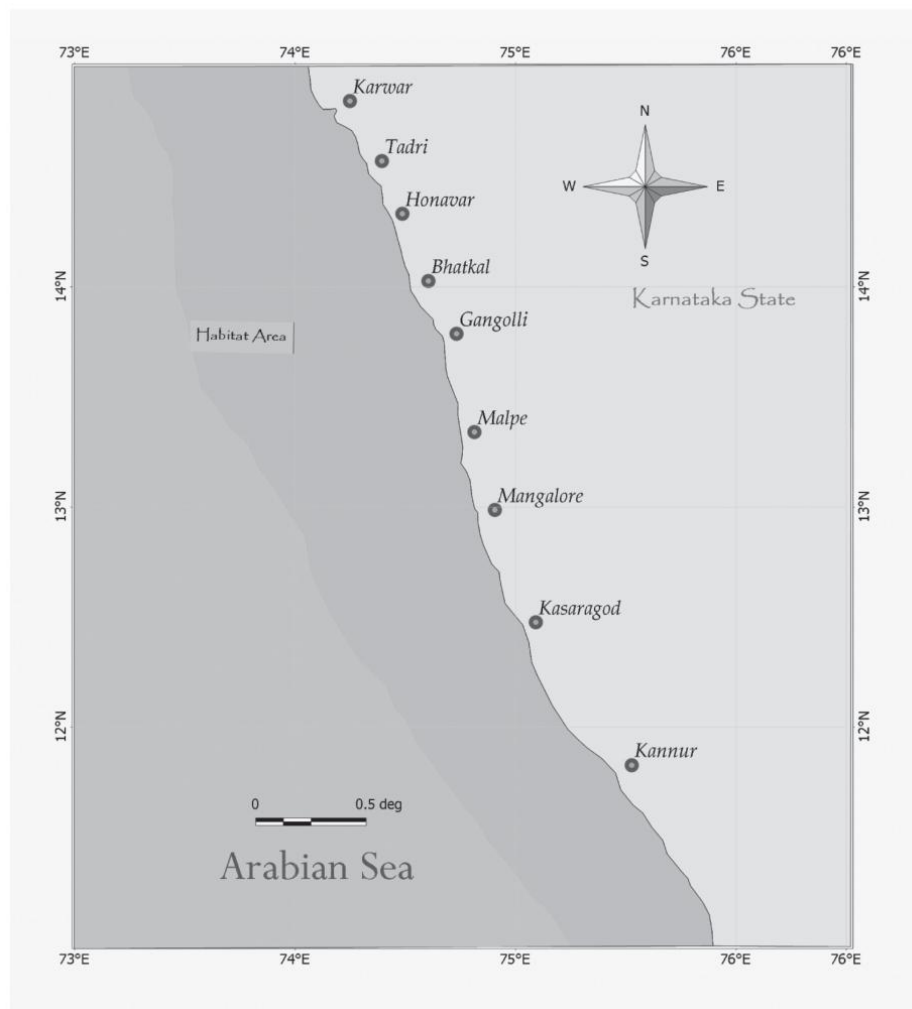
Target region boundary



Arabian Sea LME_032. Sea Around Us boundary; WGS 84 geodesic area approximately 3.93 million km². Orange line locates the Karnataka coast near the study; it is not a digitized study polygon. | [Boundary source](#) | [Overlap evidence](#)

Study area in the article

Study boundary for the selected model



Mohamed et al. (2008), Habitat Area figure, printed p. 8 / PDF p. 18. Karnataka shelf and parts of slope; source area 27,000 km². | [Source PDF](#)